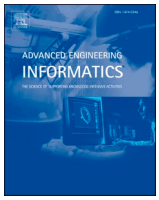
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A transformer-based surrogate modeling strategy for tunnel digital twin in full-field displacement prediction under adjacent tunnel construction

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ABSTRACT

Accurately predicting ground settlement and tunnel lining deformation is essential for ensuring the safety and efficiency of shield tunnel projects, especially in urban areas affected by adjacent excavations. Rapid and reliable forecasts of tunnel behavior support proactive design decisions and risk mitigation. However, most existing surrogate models offer limited predictive resolution and accuracy, as they predict responses only at sparse monitoring points and ignore spatial information, which restricts their applicability in building comprehensive digital twin (DT) systems for full-field simulation. To address these limitations, this study proposes a DT framework for predicting ground settlement and lining deformation in shield tunnels affected by adjacent excavation. A hybrid surrogate model, called the Dual-Fusion Swin-Table Transformer Network (DF-STTN), is developed by combining Swin Transformer and TabTransformer to handle spatial and non-spatial inputs. To enhance multimodal interaction, a dual-stage fusion strategy is applied. The model is trained on high-fidelity finite element data, then updated and validated using results from laboratory-scale tunnel excavation tests. Results demonstrate that the inclusion of spatial information improves prediction accuracy, with notable improvements achieved when three measurement points of the tunnel are used. DF-STTN outperforms conventional convolution-based and single-fusion models in both ground settlement and lining deformation tasks. The proposed framework achieves high predictive accuracy and generalization capability for early-stage tunnel design and excavation risk assessment.

1. Introduction

Urban development has led to a significant increase in underground infrastructure projects, particularly in densely populated cities where space is limited. One common engineering challenge in the design phase is planning new tunnel construction near existing operational tunnels, where the impact of excavation-induced ground movements must be carefully assessed in advance. These interactions often lead to ground settlement and structural deformation of the existing tunnel lining, which may compromise structural integrity and operational safety [25,42]. Moreover, due to geological variability, complex soil-structure interaction, and evolving excavation stages, conventional deterministic models and static empirical charts often fail to capture the full range of possible deformation behaviors during adjacent tunneling[31]. Reliable prediction of such responses during the early design stage is therefore critical to avoid costly remediation, ensure compliance with safety standards, and support decision-making in the planning and design.

Recent advancements in computational modeling and sensing technologies have introduced the concept of a digital twin (DT) [10,18,19,36]. A DT is a virtual replica of a physical system, capable of dynamically utilizing various data sources to simulate structural behavior under different scenarios[2,6]. In tunneling applications, a DT can be used during the design stage to evaluate the potential impact of sequential excavation on nearby structures, allowing engineers to proactively assess risk and optimize design strategies. This approach enhances the ability to test multiple design alternatives virtually, reducing reliance on conservative assumptions and overdesign. Despite its potential, the application of DTs in the design of underground infrastructure presents several challenges. High-fidelity numerical simulations, such as finite element (FE) models, are commonly used to represent soil–structure interaction but are computationally intensive and not easily adaptable when new data becomes available[3,4,7,34].

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Additionally, the behavior of tunnels under adjacent excavation is highly nonlinear and dependent on spatial–temporal factors, making it difficult for traditional models to generalize across different construction scenarios[5]. This gap between simulation and reality highlights the need for lightweight and effective models that can combine simulation accuracy with speed to support rapid exploration of design options.

Data-driven models, particularly surrogate models, have gained attention in this regard. These models approximate the behavior of more complex simulations by learning from precomputed results or experimental data, enabling faster prediction with minimal loss of accuracy [8,9,43,44]. In tunnel engineering, surrogate modeling approximates the nonlinear mapping between excavation/tunnel parameters and deformation responses, thereby reducing reliance on costly FE simulations. This approach follows the fundamental principle of function approximation, where machine learning models serve as universal approximators of complex soil–structure interaction functions that are otherwise captured through high-fidelity but computationally expensive FE simulations. There are various surrogate modeling approaches available, such as response surface methodology [24], Kriging [32], regression methods [1,16,46], polynomial expansions [15], and proper orthogonal decomposition [13,41]. In general, these surrogate modeling methods perform effectively when the prediction task involves only a few output variables, such as surface settlements at selected locations or deformations at key points around the tunnel lining [22]. However, the key challenge in the context of DTs for adjacent tunneling is that reliable design requires capturing full-field responses across many spatial positions, excavation stages, and even different types of outputs (e.g., ground settlement, tunnel lining deformation). Unlike point-based predictions, which only estimate behavior at selected monitoring locations, full-field prediction demands a model that can represent the continuous deformation field of the ground and lining. In this case, traditional surrogate modeling approaches reveal clear shortcomings. First, they typically scale poorly because their formulations (e.g., polynomial expansions, response surfaces, or regression-based methods) become unstable or computationally prohibitive as the dimensionality of the problem increases. Second, they often rely on simplified or limited input features, which restricts their ability to incorporate diverse information such as soil heterogeneity, construction parameters, and spatial geometry. Third, they struggle to generalize beyond the conditions represented in the training data, making them sensitive to changes in geology, tunnel geometry, or construction sequence [2,3,4,6,22]. To address these limitations, more advanced surrogate modeling techniques are needed. Such models would enable the development of comprehensive and adaptive DT frameworks capable of delivering full-field predictions under varying scenarios.

Multimodal data fusion has emerged as a promising approach for enhancing surrogate modeling in complex engineering tasks, as it enables the integration of diverse data types, such as spatial coordinates, tabular features, and sequential information, into a unified learning framework [11]. By combining different modalities, models can better capture the intricate relationships between excavation parameters, geological conditions, and system responses, ultimately improving predictive performance and generalization. For example, Luo et al. [26] proposed a visual navigation framework combining Multimodal Graph Learning and Action Boost Memory to help agents navigate using only visual input. By dynamically modeling spatial, semantic, and historical information, the method significantly improves navigation success and efficiency in unseen environments. Yang et al. [38,39] proposes a Multimodal Knowledge-Guided Model (MKGM) for the joint recognition of bridge defects and structural information by integrating visual features, knowledge graphs, and label co-occurrence through a dual-graph convolutional network and MambaVision. By capturing hierarchical semantic associations among defects, components, and structural regions, the model achieves superior performance. These examples collectively highlight the growing potential of multimodal data fusion to improve model robustness, accuracy, and applicability in a wide range of engineering contexts.

In this paper, the DT framework is designed to provide a lightweight yet accurate virtual replica of adjacent tunneling processes, with the specific goal of predicting full-field deformation of an existing tunnel during new excavation. The Dual-Fusion Swin-Table Transformer Network (DF-STTN) serves as the methodological core of the proposed DT framework. This capability is realized through a hybrid Transformer-based surrogate modeling architecture that integrates both spatial and tabular modalities [20,33]. Transformers, originally developed for natural language processing, are chosen in this work because of their theoretical strength in capturing long-range dependencies through self-attention mechanisms. This property is particularly relevant for tunneling problems, where localized excavation can induce deformation at distant parts of the tunnel and ground surface. In contrast to convolution-based networks that emphasize local features, attention-based models dynamically assign importance weights across the entire input domain, making them well-suited for geotechnical systems characterized by both local variability and global interactions[21,35,45]. Spatio-temporal models (e.g., recurrent networks, ConvLSTMs) are effective at learning sequential dynamics but often face challenges in scalability and capturing long-range spatial dependencies, which are essential in adjacent tunneling where excavation at one location can influence deformation far away [40]. Graph-based models can represent irregular geometries and local connectivity, yet they typically require complex graph construction and incur high computational costs when extended to dense, full-field predictions [14]. Moreover, adjacent tunneling problems involve two distinct types of data: (1) non-spatial/tabular data, such as soil stiffness, tunnel burial depth, and lining thickness, which describe global system properties, and (2) spatial data, such as settlement and deformation fields, which capture location-dependent responses. By explicitly designing a dual-fusion Transformer architecture, this study ensures that both modalities are processed appropriately: TabTransformer for tabular features and Swin Transformer for spatial grids, before being integrated in a complementary way. This design reflects the principle that predictive accuracy improves when models account for both global system descriptors and local spatial patterns. This makes them especially well-suited for applications involving high-dimensional, structured, or sequential data. This architecture forms the core of the proposed DT framework, supporting high-resolution, multimodal prediction of tunnel behavior.

In summary, this study proposes a DT framework tailored for the design phase of tunnel projects involving adjacent excavation. The proposed approach integrates deep learning-based surrogate modeling, feature encoding, and model updating to construct a lightweight and adaptive DT capable of simulating the deformation behavior of an existing tunnel. The proposed framework provides practical guidance for adjacent tunnel design by enabling efficient full-field prediction of deformation and supporting risk-informed engineering decision-making. Key innovations of the proposed framework include:

(1) Spatial and non-spatial information are obtained from a parametric FE model, where simulation outputs are converted into structured inputs for the surrogate model. Including spatial features, compared to counterparts using only non-spatial data, enhances the ability to capture spatial positional relationships and improves prediction accuracy.
(2) A surrogate model, i.e., Dual-Fusion Swin-Table Transformer Network (DF-STTN), is developed to integrate spatial and tabular modalities. It employs a two-stage structure to separately predict ground settlement and lining deformation, improving interpretability and accuracy. Spatial and non-spatial features are extracted using a Swin Transformer and TabTransformer, then fused via dual-stage fusion (early and late) for effective multimodal interaction.
(3) A data-driven model updating method is used to iteratively best estimate soil parameters by minimizing prediction errors against

test data. This process enables adaptive updating of the surrogate model, allowing it to generalize and remain effective under different excavation scenarios or test setups.

(4) A laboratory-scale test, emulating real tunnel projects, is conducted and then used to validate the proposed strategy of using the new surrogate model, ensuring the modeled system accurately reflects the observed tunnel behavior.

2. Methodology

2.1. Overview of the method

This study proposes a DT framework for tunnel behavior prediction and evaluation. The DT framework is structured in four layers (see Fig. 1): (1) Physical system layer, where a scaled tunnel test system represents the real-world twin; (2) Data acquisition layer, which collects measurements such as settlement, lining deformation, and acceleration; (3) Digital modeling and data fusion layers, where numerical simulations and multimodal features are used to train the surrogate model (DF-STTN) and iteratively update soil parameters; and (4) Service and interaction layer, which enables prediction, assessment, and test setup adjustments.

Firstly, the physical twin is established using a laboratory-scale testbed designed to replicate tunnel excavation scenarios. The system includes a physical tunnel model, a data acquisition system, and a sensor network that captures tunnel responses including settlement, lining deformation, strain, soil pressure, and acceleration. These experimental measurements form the test dataset, which is used both for model validation and for updating the virtual model.

It should be clarified that, due to the data availability and setup flexibilities, a laboratory-scale setup is adopted in this study to establish the DT framework. In particular, compared to the real tunnel physical twin, the designed system offers more flexibilities of setup configurations, e.g., tunnel diameters and inter-tunnel distances, and provides more comprehensive data collection schemes in the lab. Furthermore, the proposed methodology is not limited to such lab test configuration, which can be further extended and applicable to real tunnels in full-field applications, if available.

Secondly, a high-fidelity FE Model is constructed to simulate tunnel responses under varying design and soil conditions. This includes parametric modeling where multiple input variables (e.g., soil properties, tunnel geometry) are systematically varied, verification of simulation results, and generation of a comprehensive dataset comprising over 1000 samples. Key outputs from the simulations, such as ground settlement and lining deformation, are used to build the training dataset for the surrogate model.

Thirdly, a deep learning-based surrogate model is developed to replace the computationally intensive FE model simulations. This model includes both non-spatial (design parameters and soil properties) and spatial (a distance matrix representing spatial relationships between tunnel segments and soil) input features. To maintain accuracy under real-world conditions, a model updating module is integrated. It uses

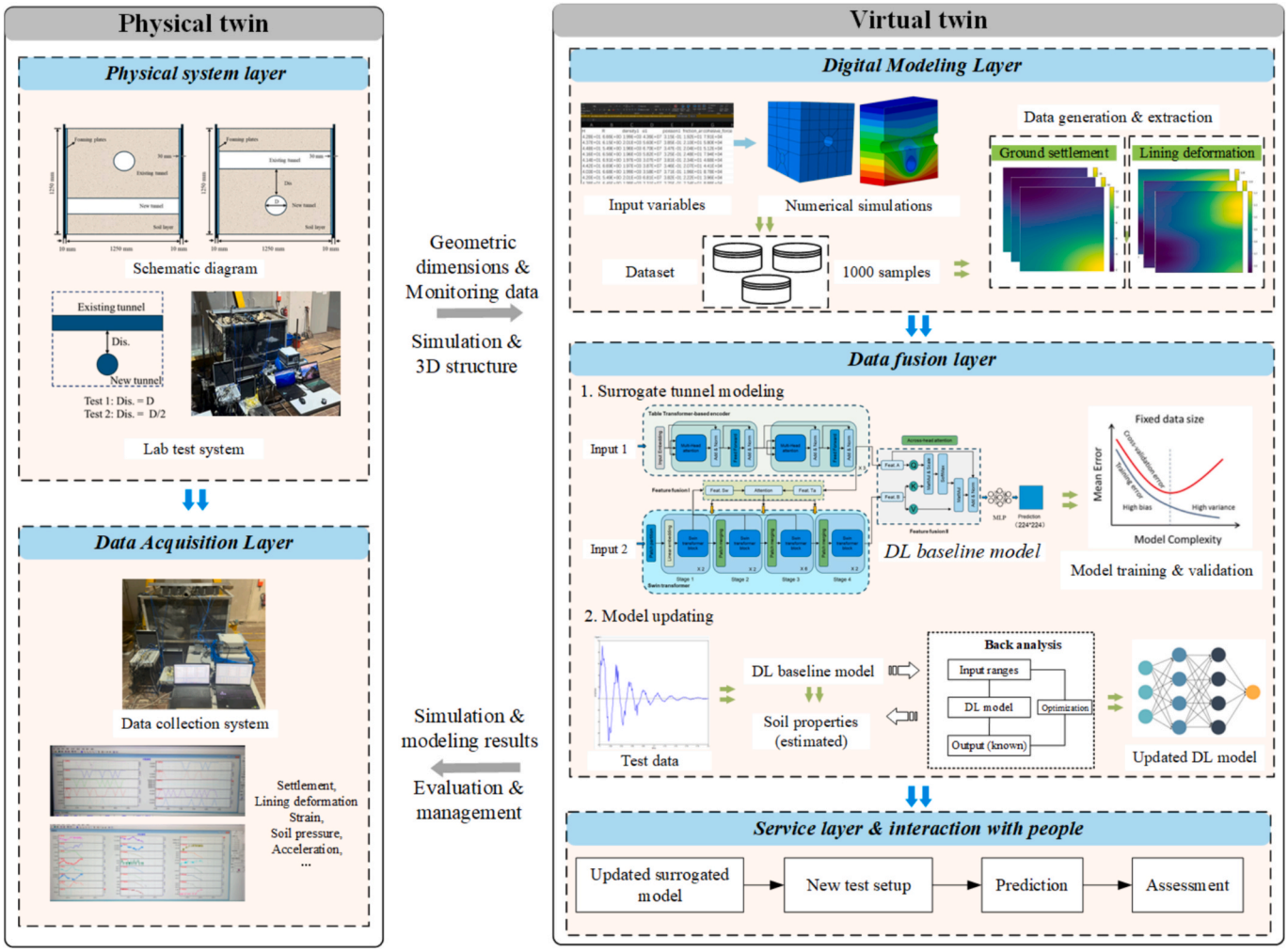


Fig. 1. The framework of the proposed method.

test data from the physical layer to iteratively adjust key model parameters, especially soil properties, through optimization techniques.

Finally, the updated model is used to provide rapid predictions for new test scenarios or design variations. This integrated DT framework enables accurate, tunnel behavior prediction and model updating by seamlessly combining physical testing, numerical simulation, and deep learning.

2.2. Data feature extraction

The input data used for surrogate model training includes both non-spatial and spatial information extracted from the FE simulation results. Two distinct types of spatial information, ground settlement and lining deformation, are extracted separately to construct two surrogate models, each tailored to its respective output type. Two types of feature information are illustrated as follows.

Non-spatial information: Non-spatial features consist of scalar input parameters that describe the tunnel design and soil conditions. These include soil mechanical properties (e.g., stiffness, friction angle, cohesion), tunnel geometric parameters, and construction-related inputs such as burial depth or cover thickness. These parameters are directly extracted from the FE model input.

Spatial information: Spatial information captures the geometric and positional context of deformation responses in the tunnel system. It is essential for learning location-dependent patterns in both ground and lining behavior. Fig. 2 illustrates the procedures of spatial information extraction.

Table 1 summarizes input and output variables for the surrogate model. Two distinct types of spatial data are extracted from the FE model outputs: ground settlement features and lining deformation features. For each case, the input is structured as a 224 × 224 × 3 grid, where each spatial point includes three components representing relative distances between the surface (or lining) grid points and predefined tunnel measurement points. The 224 × 224 grid resolution was chosen as a balance between spatial detail and computational efficiency. Correspondingly, each case has a single-channel 224 × 224 target output, which is the interpolated deformation field (ground settlement or lining deformation) derived from the FE simulation. This output serves as the prediction target for the surrogate model.

(a) Ground settlement spatial information

Ground settlement features represent the distance features on the surface above the tunnel, influenced by tunnel geometry and location.

Table 1
Summary of input and output variables for the surrogate model.

Category	Variable type	Description	Format/ Dimension
Non-spatial inputs 1×7 tensor	Soil stiffness	Elastic modulus of soil strata	Scalar
	Density	Density of soil strata	Scalar
	Poisson's ratio	Poisson's ratio of soil strata	Scalar
	Friction angle	Internal friction angle of soil	Scalar
	Cohesion	Cohesion parameter of soil	Scalar
	Tunnel diameter	Tunnel geometry, such as tunnel diameter, lining thickness, etc.	Scalar
	Tunnel depth	Parameters related to excavation/construction process	Scalar
Spatial inputs	Ground settlement grid	Relative distances of ground surface points to reference tunnel points	224 × 224 × 3 tensor
	Lining deformation grid	Relative distances of lining points to reference tunnel points	224 × 224 × 3 tensor
Outputs	Ground settlement	Interpolated settlement deformation field	224 × 224 × 1 tensor (single map)
	Lining deformation	Interpolated lining deformation field	224 × 224 × 1 tensor (single map)

These features are organized on a uniform grid to reflect spatial variations in surface location. To prepare this data for model input, the following preprocessing steps are applied to extract and standardize the spatial information:

Step 1: Grid extraction: The ground surface information from the FE model is extracted and projected onto a surface grid. Due to symmetry, only half the domain is considered, and the data is mirrored as necessary.

Step 2: Interpolation: Displacement values from the FE mesh are interpolated onto a uniform grid of size 224 × 224 to ensure consistency across samples. A 2D deformation map $\boldsymbol{d}_{zd}$ representing the vertical settlement distribution on the ground surface. This serves as the prediction target for the surrogate model.

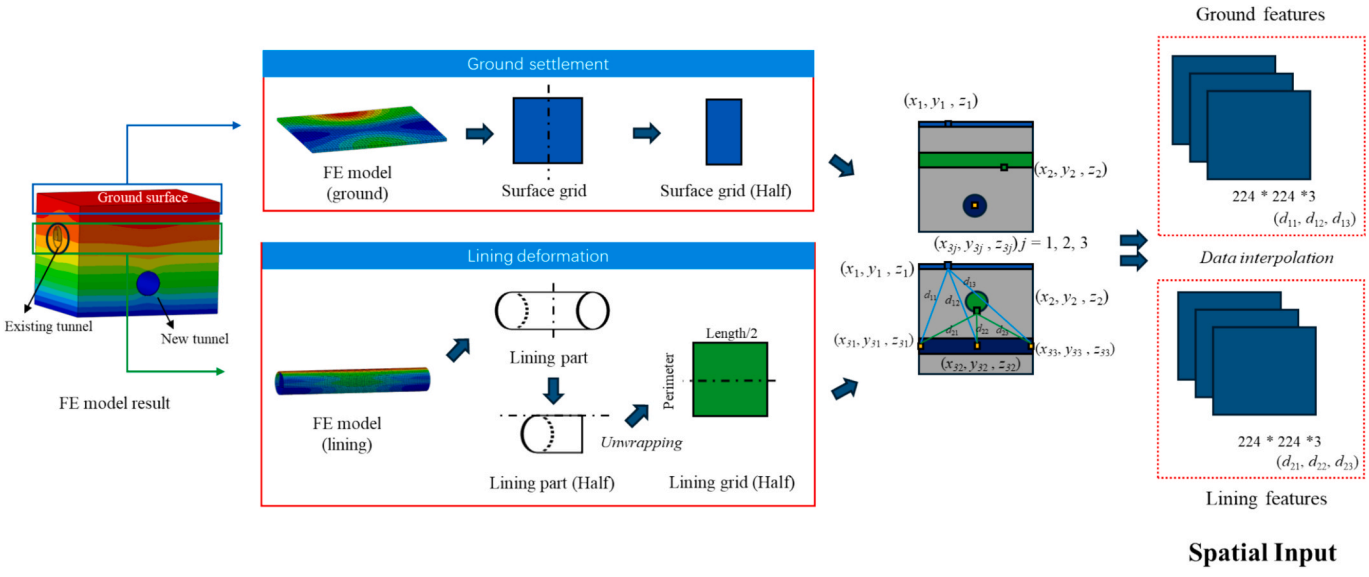


Fig. 2. Spatial information extraction.

Step 3: Spatial feature components: Each grid point on the 224 × 224 ground grid is paired with spatial descriptors that encode its relative position to measurement points of the new tunnel. These spatial features help the model understand how ground-level deformation correlates with the tunnel structure below. Specifically, each grid point is associated with a tuple:

$$(d_1, d_2, d_3) \tag{1}$$

where $d_j = \sqrt{(x_{1i} - x_{3j})^2 + \left(y_{1i} - y_{3j}\right)^2 + (z_{1i} - z_{3j})^2}$ is the Euclidean distance from the grid point (x_{1i}, y_{1i}, z_{1i}) to the jth tunnel measurement point, for $j = 1, 2, 3$.

(b) Lining deformation spatial information

Lining deformation features capture distance features on the outer surface of the tunnel lining. Due to the cylindrical geometry, a special coordinate transformation is required to unwrap the surface into a planar form suitable for modeling. Fig. 3 shows the procedures of spatial information extraction of the lining deformation.

Step 1: Geometry Unwrapping: The cylindrical tunnel lining is unwrapped into a rectangular representation, where the horizontal axis corresponds to the arc length around the tunnel circumference, and the vertical axis represents the longitudinal tunnel axis.

Step 2: Coordinate Transformation: To unwrap the cylindrical tunnel lining onto a 2D plane, we first compute the arc length between a reference vertex (typically located at the top of the tunnel) and each node along the outer perimeter of the tunnel. As shown in Fig. 3, this step is calculated by identifying the angle γ between the reference direction (from the center to the vertex) and the line connecting the center to the target node (x_c, y_c, z_c).

Given the tunnel outer radius R, the arc length for node i is calculated as:

$$arc_length_i = R \cdot \gamma \tag{2}$$

where the angle γ is obtained using trigonometric geometry in the tunnel cross-section. This *arc_length* represents the vertical coordinate of the node in the unwrapped 2D plane, with the horizontal coordinate being the value x_i in the tunnel longitudinal axis.

Step 3: Mesh Inconsistency Adjustment: Due to geometric mismatches between the high-fidelity FE model and the simplified tunnel representation, direct alignment between mesh nodes cannot always be achieved. To address this, a local neighborhood interpolation strategy is applied to remap the high-resolution FE data onto a uniform grid in the unwrapped 2D plane. This ensures consistent spatial correspondence between all samples. A 2D deformation map $\boldsymbol{d}_{zl}$ representing the lining deformation distribution of the tunnel lining. This serves as the prediction target for the surrogate model.

Step 4: Spatial Feature Components: Although the mesh has been interpolated into a 2D format for consistency, this interpolation alters the original node locations. Therefore, to preserve spatial awareness for modeling, we recover the 3D coordinates (x, y, z) of each grid point on the cylindrical tunnel surface using the tunnel outer radius R and the angle θ derived from *arc_length*. This gives the full 3D position of the point:

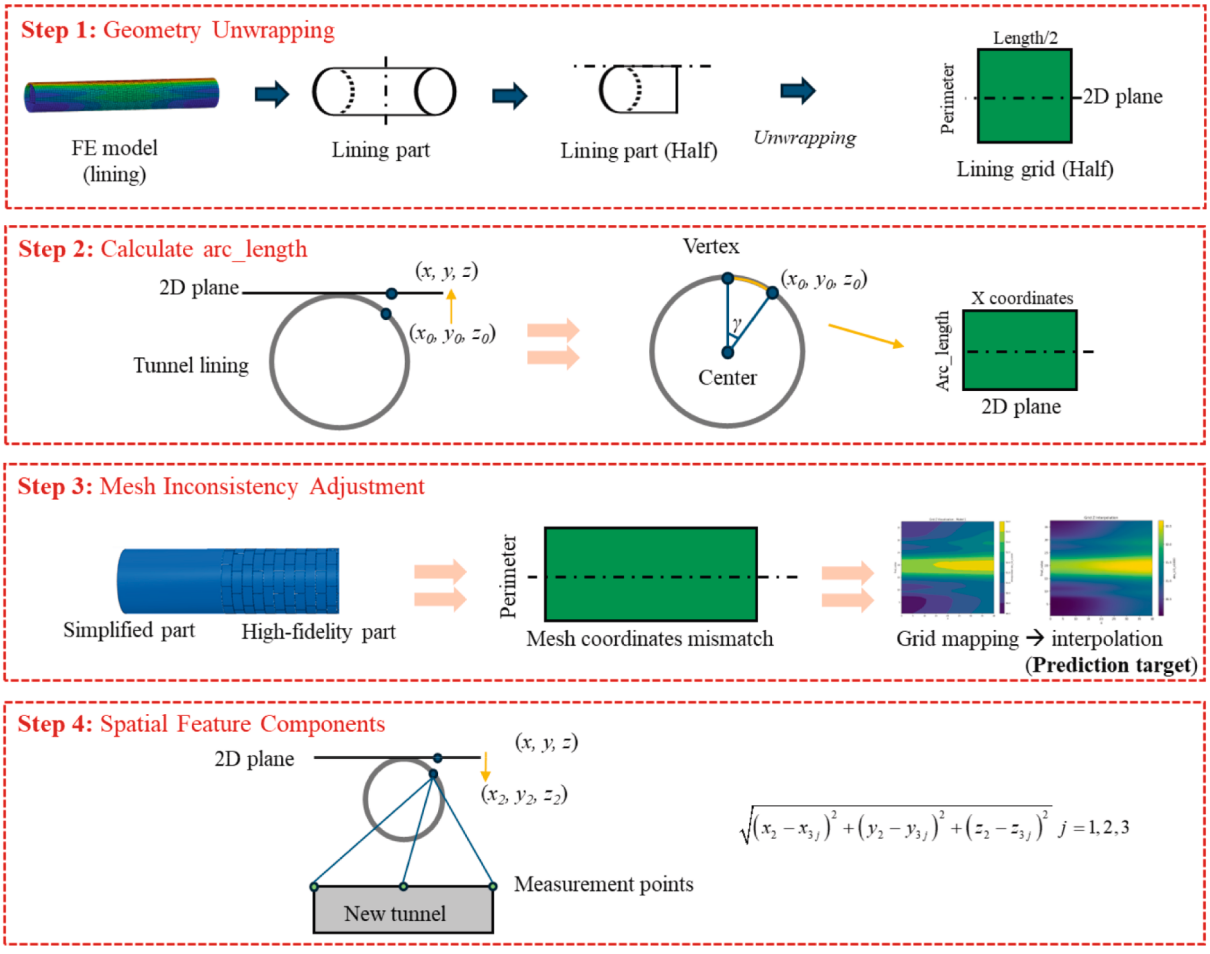


Fig. 3. Procedures of spatial information extraction of the lining deformation.

$(x_2,\ y_2,\ z_2)$

Then, for each measurement point $P_j = \left(x_{3j},\ y_{3j},\ z_{3j}\right)$ on the tunnel, spatial information on the 224 × 224 × 3 lining grid is calculated by:

$$d_j = \sqrt{\left(x_2 - x_{3j}\right)^2 + \left(y_2 - y_{3j}\right)^2 + \left(z_2 - z_{3j}\right)^2}\ j = 1, 2, 3 \tag{3}$$

where these distances (d_1, d_2, d_3) represent the spatial relationships between each lining grid point and the tunnel measurement points.

2.3. Surrogate modeling method

To approximate the outputs of computationally expensive FE model simulations, a surrogate model that combines spatial and tabular modalities through the DF-STTN is developed (see Fig. 4). The proposed DF-STTN is grounded in the fundamental principle of function approximation, where deep neural networks serve as universal approximators that can map complex nonlinear relationships between input parameters (e.g., soil stiffness, tunnel geometry, depth) and output deformation responses. Its domain-specific superiority lies in its alignment with physical deformation behavior. In the context of adjacent tunneling, these nonlinear mappings represent the physical relationship between excavation-induced stress redistribution and resulting ground and lining deformation. The model's dual-modal fusion mechanism enables simultaneous learning of global structural effects, captured through features like tunnel geometry, depth, and soil properties, and localized spatial interactions, represented by 2D spatial grids that reflect how deformation propagates around tunnels and interfaces. Additionally, the cross-modal fusion module captures correlations between input parameters and localized deformation responses, consistent with established geotechnical principles where stiffer soils or deeper tunnels constrain settlement.

The model consists of three major components: (1) a Table Transformer encoder for non-spatial features, (2) a Swin Transformer encoder for spatial features, and (3) a cross-attention module for multimodal fusion. The goal is to learn a mapping:

$$f_\theta : \left(\boldsymbol{X}^{\text{non-spatial}},\ \boldsymbol{X}^{\text{spatial}}\right) \rightarrow \mathbf{y} \tag{4}$$

where $\boldsymbol{X}^{\text{spatial}} \in \mathbb{R}^{H\times W\times C}$ is the spatial input tensor, $\boldsymbol{X}^{nonspatial} \in \mathbb{R}^{d}$ is the vector of non-spatial input, and $\mathbf{y}$ is the predicted deformation response (e.g., ground settlement and lining deformation).

(a) Table Transformer for non-spatial feature encoding.

The non-spatial input $\boldsymbol{X}^{nonspatial} \in \mathbb{R}^{d}$ is embedded as a token sequence[27]:

$$\boldsymbol{T}^{(0)} = [\boldsymbol{t}_1,\ \boldsymbol{t}_2, \cdots, \boldsymbol{t}_m] \tag{5}$$

This sequence is then passed through a series of Transformer encoder layers. Each layer is composed of two primary sub-blocks: a multi-head self-attention mechanism (MHSA) and a feed-forward network (FFN).

In MHSA, the token sequence $\mathbf{T}^{(l)}$ at layer l is processed as follows:

$$\text{MHSA}\left(\boldsymbol{T}^{(l)}\right) = \text{Concat}(\text{head}_1,\ \cdots,\ \text{head}_h)\boldsymbol{W}^O \tag{6}$$

where each attention head is computed by:

$$\text{head}_i = \text{Attention}\left(\boldsymbol{T}^{(l)}\boldsymbol{W}_i^Q,\ \boldsymbol{T}^{(l)}\boldsymbol{W}_i^K,\ \boldsymbol{T}^{(l)}\boldsymbol{W}_i^V\right) \tag{7}$$

The output of the attention block is then passed through a FFN of the form:

$$\text{FFN}(\boldsymbol{t}) = \sigma(\boldsymbol{t}\boldsymbol{W}_1 + \boldsymbol{b}_1)\boldsymbol{W}_2 + \boldsymbol{b}_2 \tag{8}$$

where $\sigma(\cdot)$ denotes a non-linear activation function (i.e., ReLU), and $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{b}_1$, and $\mathbf{b}_2$ are trainable parameters.

Each sub-block is followed by layer normalization and a residual connection, which help keep the training process stable and ensure smoother gradient flow across the transformer layers.[33].

(b) Swin Transformer for spatial feature encoding.

Let the spatial input be divided into patches of size $P \times P$, giving $N = \frac{H}{P}\frac{W}{P}$ patches. Each patch $\boldsymbol{x}_i \in \mathbb{R}^{P^2\cdot C}$ is linearly projected into an embedding vector $\boldsymbol{z}_i \in \mathbb{R}^{d}$, forming the sequence[20]:

$$\boldsymbol{Z}^{(0)} = [\boldsymbol{z}_1,\ \boldsymbol{z}_2,\ \cdots,\ \boldsymbol{z}_N] \in \mathbb{R}^{N\times d} \tag{9}$$

The sequence is then passed through a series of Swin Transformer blocks, where each block consists of a Window-based Multi-Head Self-Attention (W-MSA) layer and a Shifted Window MSA (SW-MSA) layer in alternating fashion [23]. These attention mechanisms are used to capture both local and global dependencies across the spatial domain [38,39].

Within each attention layer, the input sequence $\mathbf{Z}^{(l)}$ is projected into three matrices: the query matrix $\mathbf{Q}$, key matrix $\mathbf{K}$, and value matrix $\mathbf{V}$, as follows:

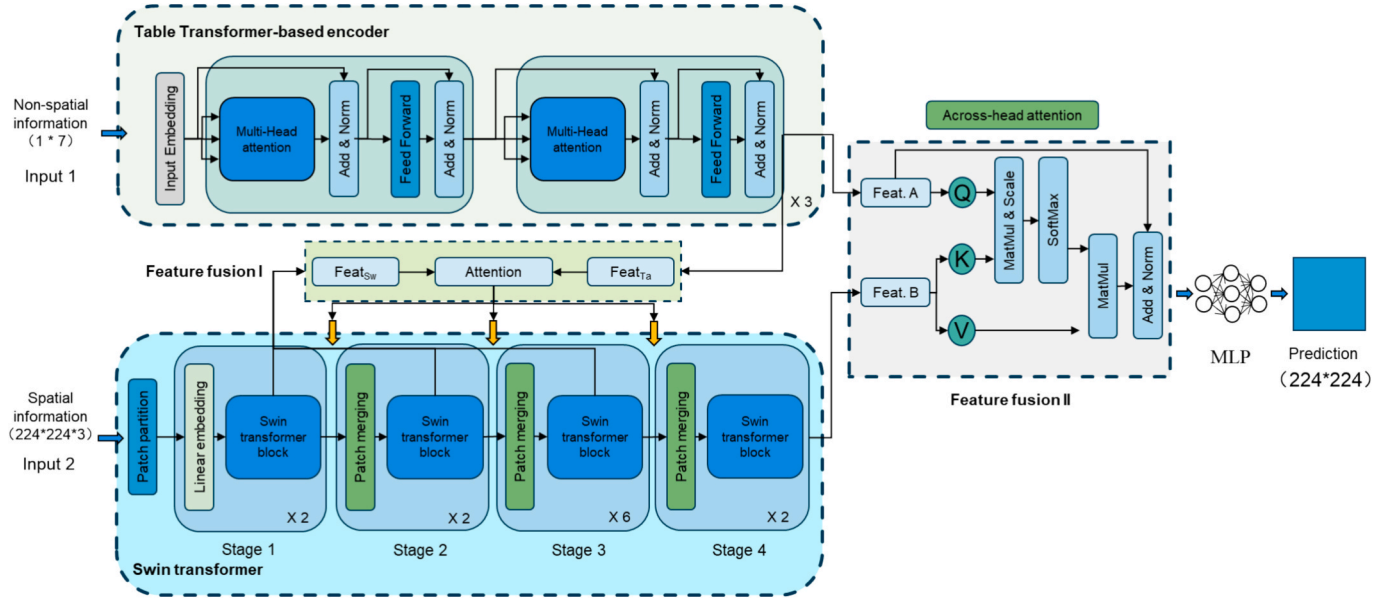


Fig. 4. DF-STTN for the surrogate model.

$$\boldsymbol{Q} = \boldsymbol{Z}^{(l)}\boldsymbol{W}^{Q}, \quad \boldsymbol{K} = \boldsymbol{Z}^{(l)}\boldsymbol{W}^{K}, \quad \boldsymbol{V} = \boldsymbol{Z}^{(l)}\boldsymbol{W}^{V} \tag{10}$$

where $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are learnable projection matrices, and d_k is the dimension of each attention head. The attention output is computed using scaled dot-product attention:

$$\text{Attention}(\boldsymbol{Q},\ \boldsymbol{K},\ \boldsymbol{V}) = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{T}}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{11}$$

To enhance cross-window communication, the Swin Transformer introduces a shifted window mechanism in alternating layers. In this approach, the partitioning of attention windows is shifted by half the window size to enable interactions between neighboring regions that would otherwise be disjoint[23].

Each stage of the Swin Transformer also includes a patch merging operation, which downsamples the spatial resolution while increasing the channel depth. Formally, this is represented as:

$$\boldsymbol{Z}^{(l+1)} = \text{Downsample}\left(\boldsymbol{Z}^{(l)}\right) \tag{12}$$

This hierarchical structure enables the model to capture multi-scale spatial features effectively, which is critical for learning localized deformation patterns in complex geometries.

(c) Cross-attention fusion

To integrate information from both spatial and non-spatial modalities, the model includes two levels of feature fusion: a late-stage fusion and an early-stage fusion, both based on attention mechanisms[12,37]. These fusion strategies allow the model to effectively combine spatial features from the Swin Transformer with non-spatial features from the Table Transformer at different stages of the network.

The late-stage fusion is designed to integrate high-level semantic representations from both modalities. Let $\boldsymbol{F}^{Sw} \in \mathbb{R}^{n\times d}$ represent the spatial features obtained from the Swin Transformer encoder, and $\boldsymbol{F}^{TA} \in \mathbb{R}^{m\times d}$ denote the non-spatial feature representations output by the Table Transformer.

The cross-attention is constructed by projecting $\boldsymbol{F}^{Sw}$ into query vectors and $\boldsymbol{F}^{TA}$ into key and value vectors using learnable weight matrices [8,9]:

$$\boldsymbol{Q} = \boldsymbol{F}^{Sw}\boldsymbol{W}^{Q}, \quad \boldsymbol{K} = \boldsymbol{F}^{TA}\boldsymbol{W}^{K}, \quad \boldsymbol{V} = \boldsymbol{F}^{TA}\boldsymbol{W}^{V} \tag{13}$$

The fused representation is then computed using scaled dot-product attention:

$$\boldsymbol{F}^{\text{fused}} = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^{T}}{\sqrt{d_k}}\right)\boldsymbol{V} \tag{14}$$

Finally, an Add & Norm step is applied by combining the original Swin features with the fused output, followed by layer normalization:

$$\boldsymbol{F}^{\text{output}} = \text{LayerNorm}\left(\boldsymbol{F}^{Sw} + \boldsymbol{F}^{\text{fused}}\right) \tag{15}$$

This fused representation $\mathbf{F}^{\text{output}}$ serves as the input to the final prediction head, enabling the model to make informed predictions based on both spatial geometry and non-spatial design or material properties.

In addition to late-stage fusion, an early-stage fusion module is also implemented to inject non-spatial context earlier in the processing pipeline. This mechanism shares a similar attention-based structure with the late-stage fusion. Intermediate spatial features $\mathbf{Feat}_{\text{Sw}}$ and non-spatial features $\mathbf{Feat}_{\text{Ta}}$ are aligned and fused through attention before being fed into the deeper stages of the Swin Transformer. This early interaction enhances the feature representation by allowing spatial encoding to adapt based on design-related cues from the beginning of the network.

In summary, these two fusion strategies enable the model to leverage complementary strengths of spatial and tabular data at multiple levels, thereby improving the overall robustness and generalization of tunnel behavior prediction.

(d) Output prediction

The final fused features $\mathbf{F}^{\text{output}}$ are passed through a multilayer perceptron layer (MLP) f_{out} to generate predictions:

$$\widehat{\boldsymbol{y}} = f_{out}(\boldsymbol{F}^{output}) \tag{16}$$

The dimensionality of the prediction $\widehat{\boldsymbol{y}}$ depends on the task at hand. For ground settlement prediction, the output takes the form $\widehat{\boldsymbol{y}} \in \mathbb{R}^{224\times224}$, representing a 2D spatial map of settlement values on the surface. Similarly, for lining deformation prediction, the output also lies in $\mathbb{R}^{224\times224}$, corresponding to the unwrapped tunnel surface grid.

To train the model, the Mean Squared Error (MSE) is used as the loss function, which penalizes deviations between predicted and actual values. The MSE loss is defined as:

$$L = \frac{1}{N}\sum_{i=1}^{N}\|\widehat{\boldsymbol{y}}_i - \boldsymbol{y}_i\|_2^2 \tag{17}$$

where N is the number of training samples, $\|\widehat{\boldsymbol{y}}_i - \boldsymbol{y}_i\|_2^2$ is the squared L2 norm (Euclidean distance squared) between prediction and ground truth for the i-th sample, $\widehat{\boldsymbol{y}}_i$ is the predicted deformation map, and $\mathbf{y}_i$ is the corresponding ground truth.

2.4. Model updating strategy

The DF-STTN is utilized to effectively capture the complex nonlinear relationships between the input features and the corresponding target outputs. Then, the model is updated to ensure generalization performance. Test data from the physical experiment is used to perform back-analysis, as outlined in Algorithm 1. The baseline model is inverted to estimate unknown soil parameters by minimizing the error between the model predictions and observed test data. A genetic algorithm (GA) is applied to identify the best-fitting parameters. In principle, other optimization algorithms (e.g., Bayesian optimization, gradient-based methods) could also be adopted within the same framework.

Algorithm 1 Model updating framework based on back analysis for soil properties.

Input:
– Test data $\boldsymbol{Y}$ test (e.g., ground settlement or lining deformation)
– Initial guess of soil parameters θ_0
– Trained surrogate model $f(\cdot;\theta)$
– Optimization algorithm (e.g., GA, Bayesian optimization)
Output:
– Estimated soil parameters θ^*
1: Initialize $\theta \leftarrow \theta_0$
2: Set convergence threshold ε and maximum iterations max_iter
3: For iteration = 1 to max_iter do
4: Predict output: $\hat{Y} \leftarrow f(\theta)$
5: Compute error: $\mathscr{L} = ||\hat{Y} - \boldsymbol{Y}_\text{test}||^2$ (i.e., MSE loss)
6: If $\mathscr{L} < \varepsilon$ then
7: Break
8: Update θ using optimization algorithm to minimize $\mathscr{L}$
9: End For
10: Return optimized parameters θ^*

The updated surrogate model is then deployed for predictive analysis. The estimated soil properties and updated conditions are fed into the surrogate model to simulate future tunnel behavior. The predicted outputs are compared against experimental results for validation, ensuring the reliability of the DT system.

3. Case study

3.1. Test introduction and data collection

The engineering site selected for this study is located along Singapore's Cross-Island Line, focusing on evaluating the effects of constructing a new tunnel for the Cross-Island Line on the existing tunnels of other MRT lines, as illustrated in Fig. 5. The existing tunnel is a single-track structure composed of circular segment linings with staggered



Fig. 5. Cross Island Line in Singapore.

joints. Its key specifications are: outer diameter of 12 m, lining thickness of 0.45 m, ring width of 2 m, and tunnel depth of 35 m. The lining is made of C50-grade concrete with an elastic modulus of 35 GPa [7]. It is assumed that a new Cross-Island Line tunnel, sharing the same structural parameters, will be constructed to underpass the existing tunnels at specific locations. The crossing angle between the new large-diameter tunnel and the existing tunnels is approximately 90 degrees.

Due to the limited availability and high cost of obtaining comprehensive field monitoring data, this study adopts a laboratory-scale testing approach to evaluate the impact of adjacent tunnel excavation. While field measurements are often sparse and subject to environmental constraints, controlled laboratory tests allow for precise monitoring and repeatable conditions, enabling systematic validation of surrogate models. However, physical model tests are also time-consuming and resource-intensive. To overcome these limitations, a hybrid strategy is employed, combining experimental data with high-fidelity FE simulations, to generate a rich dataset for training and validating the surrogate model. This integration ensures both data quantity and quality while reducing the reliance on extensive physical testing. Although the lab-scale test provides valuable insights, it still has limitations such as scale effects and simplified boundary conditions. To mitigate these issues, measures such as careful material selection, similarity criteria, and cross-verification with FE outputs are implemented to enhance the representativeness and feasibility of the experiment.

3.1.1. Lab test setup

Fig. 6 illustrates the test apparatus, which is composed of four main components: a model box, an existing tunnel model, a tunnelling system, and a data acquisition system. The similarity coefficient for the geometry of the model is selected as $C_L = 48$. The outer diameter of the tunnel lining was 250 mm. The dimension of the soil mass (length × width × height) was 1250 mm × 1250 mm × 1250 mm. The PE pipe has a Young's modulus of 1.4 GPa, while the concrete lining has a Young's modulus of 35 GPa, resulting in a modulus scale factor of 25. Based on

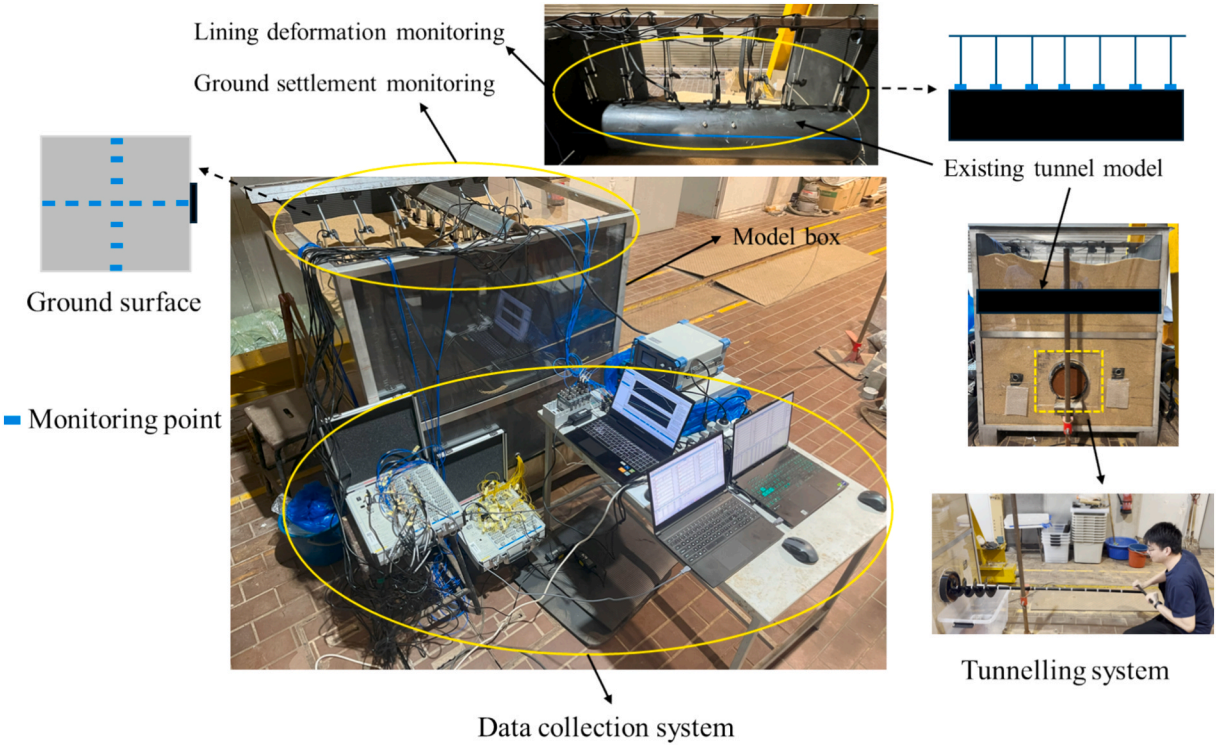


Fig. 6. Test apparatus.

this modulus scale factor and the geometric scaling, the remaining scale factors employed in the model test are derived through similitude analysis, as summarized in Table 2.

Linear Variable Differential Transformers (LVDT) were used to monitor ground settlement and lining deformation, with the sensors evenly spaced throughout the measurement area (see Fig. 6). As shown in Fig. 7, three tests were conducted with different tunnel spacings and radiuses to analyze ground and lining deformation response.

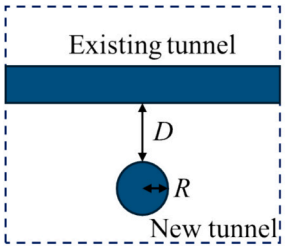
The tunnelling system comprised a tunnel lining (same as the existing tunnel), a spiral blade for excavation, and a tunnelling platform. During the excavation process, the tunnel was excavated in 13 stages, with each step advancing 10 cm. Note that the next step was only carried out once the monitoring data from all sensors had stabilized.

3.1.2. FE modelling

Fig. 8 presents a schematic of the FE model, which has dimensions of 80 m (width) × 80 m (length) × 70 m (height) to minimize boundary effects. The model details can be found in Chang et al. [7].

The soil properties listed in Table 3 were derived from laboratory tests on sand specimens and are used to model soil behavior in the FE simulations. The parameters have the following meanings: Density (D) reflects the mass per unit volume of the soil and varies with compaction state; Elastic modulus (E) represents soil stiffness and controls deformability, directly influencing ground settlement and lining response; Poisson's ratio (ν) describes the lateral strain response under axial loading, affecting soil–structure interaction; Friction angle (φ) is a measure of shear strength and governs excavation stability; and Cohesion (c), though typically small for sand, accounts for minor bonding effects observed in laboratory-prepared specimens. The variations in these properties correspond to differences in compaction and stress conditions during testing, ensuring that the simulation inputs reflect realistic soil behavior. The soil part is modeled as a homogeneous and isotropic material adhering to Mohr-Coulomb failure model. In this study, soil stratification was not explicitly considered because the laboratory soil was uniform sand. This assumption provides a reasonable simplification for the baseline model. The average values of the soil properties in Table 3 are chosen as follows: $D = 2100$ g/cm^3 (Density), $E = 60$ MPa (Elastic modulus), $\nu = 0.3$ (Poisson's ratio), $\varphi = 25^\circ$ (Friction angle), $c = 1.005$ kPa (Cohesion). For design parameters, $TD = 45$ m (Depth of the new tunnel), and $TDi = 12$ m (Diameter of the new tunnel). These parameters were used as input for the Mohr-Coulomb failure model. All other settings were kept the same as those in Test 1. The detailed simulation procedures for the new tunnel excavation are summarized as follows:

(1) The initial step comprises two sequential phases. First, the tunnel model is deactivated to allow the surrounding soil to reach a natural equilibrium state. Subsequently, the existing tunnel is introduced by activating its lining elements. The system, comprising both soil and tunnel, is then allowed to continue settling until a stable equilibrium is achieved. The displacement field obtained at this stage is recorded as the initial condition and will be subtracted from the subsequent results to accurately simulate the excavation-induced deformations.
(2) The soil elements ahead of the tunnel face are deactivated to simulate the removal of ground during shield advancement. Face pressure is then applied at the excavation front to maintain stability. Subsequently, the shield tail region is activated, followed by the simulation of tail grouting. Finally, the tunnel lining elements are activated to provide structural support. During the simulation, soil settlement and tunnel lining deformation are monitored to assess the structural stability of the tunnel.



	Distance (D)	Radius (R)
Test 1	$2R$	R
Test 2	R	R
Test 3	$2R$	$0.8R$

Fig. 7. Experimental condition.

Figs. 9 and 10 present a comparison between the numerical results and the measured data at corresponding locations. The numerical model demonstrates a similar trend to the experimental observations, suggesting that the finite difference model performs reasonably well. However, noticeable discrepancies remain, highlighting the presence of significant residuals. This indicates a need for further refinement of the soil property parameters to better represent the actual ground conditions.

3.2. Surrogate model of ground settlement

3.2.1. Surrogate model development

Seven input variables of numerical simulations were set to 1000 levels each, including five parameters of the soil part and the new tunnel depth and diameter. By substituting the sample data obtained from the LHS experimental design into the parametric FE model for calculation, 1000 effective sample datasets can be obtained. The dataset is then shuffled and split into two parts: 800 samples are designated as the training set, while the remaining 200 samples are used as the validation and test set. The input data used for surrogate model training includes both non-spatial (1 × 7) and spatial (224 × 224 × 3) information. Ground settlement fields with 224 × 224 derived from the FE simulation serve as the prediction targets for the surrogate model. Table 4 shows the hyperparameters of DF-STTN. The selection of these hyperparameters was based on a combination of empirical tuning and reference to established practices in Transformer-based models [23,27].

The effect of training sample size on the predictive performance of the surrogate model was assessed by varying the number of training samples from 100 to 800. As presented in Table 5, the MSE exhibited a decreasing trend with the increase in sample size. A substantial reduction in MSE was observed between 100 and 500 samples, dropping from 7.5367 to 1.1755, indicating a significant improvement in model accuracy during the early stages of training. Beyond 500 samples, the rate of improvement diminished, with the MSE gradually decreasing to 0.8565 at 800 samples. This trend indicates that although increasing the amount of training data further improves model performance, the incremental improvements become progressively smaller. Therefore, 700 samples can be considered a cost-effective threshold, balancing model accuracy with computational and data collection costs.

To evaluate the influence of spatial information on model performance, different numbers of measurement points along the tunnel were included in the input data. The configurations are illustrated in Fig. 11, where spatial points were progressively added along the tunnel

Table 2
Scale factors of the tunnel testbed.

Parameter dimension	Length L	Stiffness FL^{-2}	Force F	Mass FL^{-3}	Stress FL^{-2}	Strain 1	Moment FL
Scale factor	48	25	4.0E + 04	0.625	25	1.0	1.6E + 06

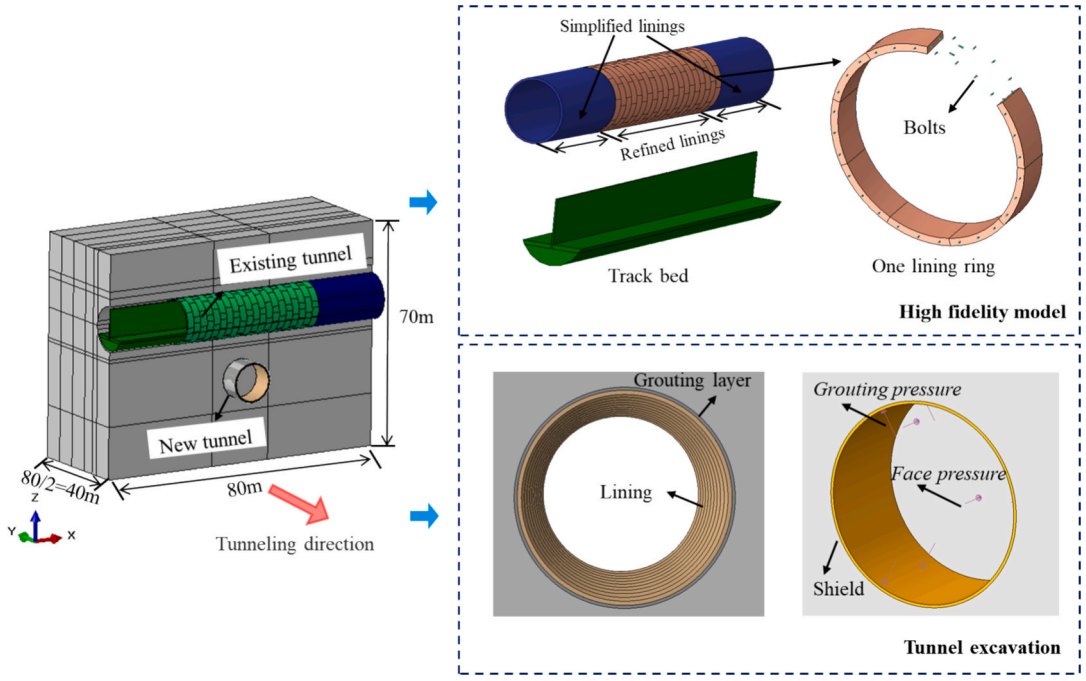


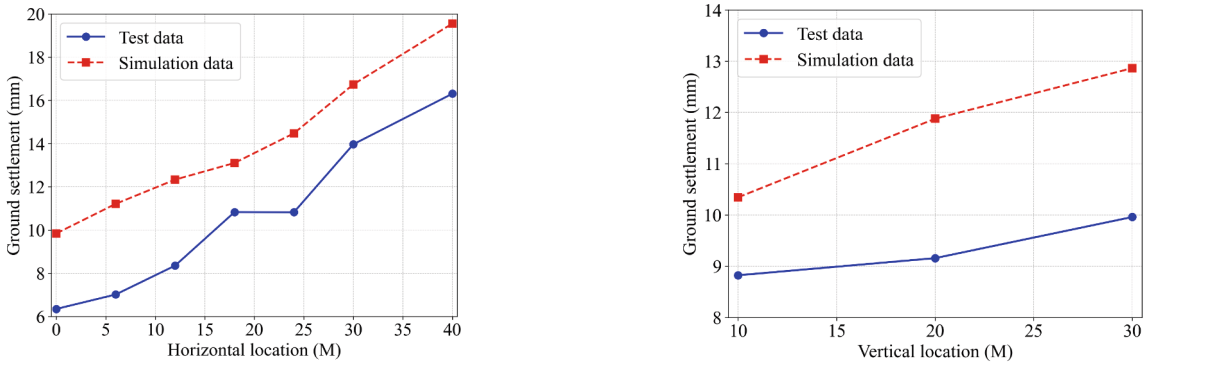
Fig. 8. The FE model in Abaqus.

Table 3
Lower/upper bounds of input variables of numerical simulations.

Parameter/unit	Range
Soil	
Density (g/cm^3)	1950–2250
Elastic modulus (MPa)	40–80
Poisson's ratio	0.2–0.4
Friction angle (o)	10–40
Cohesion (kPa)	0.01–2
Design parameters	
Depth of the new tunnel (m)	35–50
Diameter of the new tunnel (m)	10–14

alignment to provide localized observations of ground settlement. As shown in Table 6, the inclusion of spatial points consistently improved the predictive accuracy of the surrogate model across all evaluation metrics. Notably, the model with no spatial input exhibited the highest MSE (1.7626), RMSE (1.3276), MAE (1.4969), and MAPE (1.7023), along with the lowest R^2 value (0.9704). With the addition of even a single spatial point, substantial improvements were observed, reducing the MSE to 1.2048 and increasing the R^2 to 0.9879. The performance continued to improve as more spatial points were added, with the largest relative gains observed up to three points. For example, the configuration with three points achieved an MSE of 0.8565 and an R^2 of 0.9898, indicating a strong fit between predictions and actual ground settlement values. Beyond three points, the improvements became more modest, suggesting diminishing returns with additional points. These results imply that incorporating a small number of spatial points (e.g., three) can already provide high predictive accuracy while maintaining computational efficiency. This highlights that even limited spatial information can meaningfully enhance the accuracy of full-field settlement prediction.

To assess the effectiveness of different surrogate modeling approaches, seven networks were evaluated on the ground settlement prediction task. For CNN-FCNN, a Convolutional Neural Network (CNN) is used to extract spatial features, while a Fully Connected Neural Network (FCNN) processes the non-spatial tabular input[28,30]. The features from both branches are concatenated and passed through fully connected layers to generate the final prediction. ResNet-FCNN replaces the CNN encoder with a ResNet-18 backbone to extract spatial features, while retaining the FCNN for processing non-spatial inputs [17]. ResNet-FiLM-I integrates spatial and non-spatial information through a Feature-



(a) Along the excavation direction (b) Perpendicular to the excavation direction

Fig. 9. Comparison of numerical and test results for ground settlement.

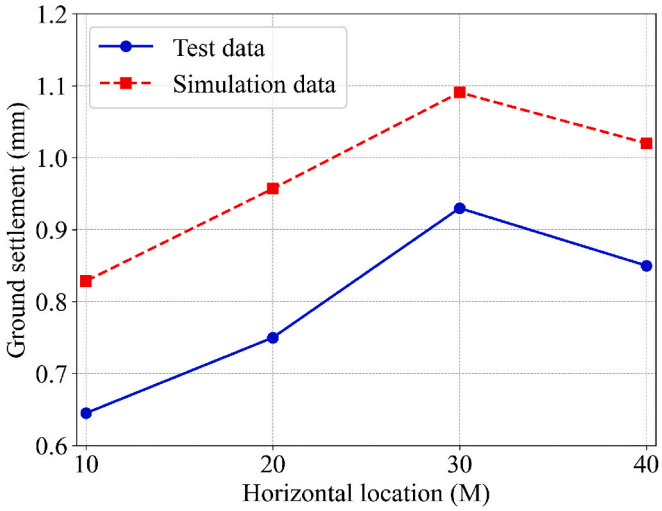


Fig. 10. Comparison of numerical and test results for lining deformation.

Table 4
Hyperparameters of DF-STTN.

Model components	Name	Value
Table Transformer encoder	Embedding dimension	128
	Number of attention heads	4
	Number of encoder layers	6
	Dimension of the feedforward network model	256
	dropout value	0.1
Swin Transformer encoder	Patch size	4
	Patch embedding dimension	96
	Depth of each Swin Transformer layer	[2, 2, 6, 2]
	Number of attention heads in different layers	[3, 6, 12, 24]
	Window size	7
	Stochastic depth rate	0.1
Cross-attention module	Node of MLP (one hidden layer)	2048

Table 5
MSE for ground settlement prediction using different training sample sizes.

Sample number	MSE
100 samples	7.5367
200 samples	3.9878
300 samples	2.5748
400 samples	1.6796
500 samples	1.1755
600 samples	0.9840
700 samples	0.8859
800 samples	0.8565

wise Linear Modulation (FiLM) mechanism[29]. A ResNet-18 backbone extracts high-level spatial features from image-based inputs. Non-spatial features are then used to modulate these spatial features via a FiLM layer, which applies learned affine transformations (scaling and shifting) to the feature maps. ResNet-FiLM-II builds upon the ResNet-FiLM architecture by integrating FiLM layers directly into the residual blocks of ResNet-18. Instead of applying FiLM modulation after feature extraction, this design fuses non-spatial features into each residual block, allowing more fine-grained and context-aware modulation throughout the spatial encoding process. Swin-T-I combines a Swin Transformer for spatial encoding and a TabTransformer for non-spatial input processing. After separate feature extraction, the outputs are flattened and concatenated, followed by a fully connected prediction head. Swin-T-II also uses Swin and TabTransformers, but fuses their outputs via an across-head cross-attention mechanism, enabling more selective integration of spatial features.

Table 7 shows model performances of seven networks. Among these, the proposed DF-STTN achieved the best overall performance, with the lowest MSE (0.8565), RMSE (0.9254), MAE (0.7102), and MAPE (0.9215), while maintaining a high R^2 score of 0.9898. The performance of ResNet-based models improved when equipped with FiLM, indicating the benefit of adaptive conditioning [29]. While the ResNet-FiLM-II achieved the highest R^2 (0.9944), its error metrics remained higher than those of the DF-STTN, suggesting a slightly inferior balance between accuracy and stability. The superiority of DF-STTN is further confirmed by the error maps shown in Table 8. These visualizations reveal that this model not only captures the overall spatial distribution of ground settlement with high fidelity but also produces the smallest deviation from ground truth across representative samples. In contrast, CNN-based models exhibit larger localized errors, especially in regions with steep gradients. These results demonstrate that integrating spatial feature extraction via Swin Transformer with tabular conditioning enables more accurate and robust predictions, making DF-STTN a strong candidate for surrogate modeling in geotechnical applications.

3.2.2. *Model updating and analysis*

To enhance the reliability of the baseline surrogate model, a model updating framework was established based on back analysis. In this study, soil parameters were chosen as the focus of the update process because they have a significant impact on ground deformation and are often highly uncertain in practice. Due to the limited availability of test data, the back analysis was conducted on a subset of specific monitoring points, as shown in Fig. 6.

GA was employed as the optimization method due to its global search capabilities and robustness against local minima. Other parameters, such as design variables (e.g., tunnel geometry and tunnel location) and spatial information, were considered known and kept constant during the optimization process to isolate the influence of soil properties. The GA settings used in this study included a population size of 50, a crossover rate of 0.8, a mutation rate of 0.1, and a maximum of 100 generations. These parameters were selected based on empirical testing to balance convergence speed and global search performance. The fitness function was defined as MSE between the predicted and observed settlement at selected monitoring points. As shown in Fig. 12, both the best and mean fitness values rapidly decreased over generations, demonstrating effective convergence of the GA. The best fitness value stabilized after approximately 20 generations, indicating that the model successfully identified a set of soil parameters yielding high predictive accuracy. The resulting predictions and corresponding true values are summarized in Fig. 12. The best individual identified by the GA corresponds to an MSE of 0.9319. The predicted values closely match the measured settlements at all selected positions, validating the effectiveness of the model updating procedure. This approach allows for the refinement of soil parameters in data-scarce conditions, thereby improving the fidelity of numerical simulations based on the trained DL model.

To evaluate the reliability of the back-analyzed soil parameters, the estimated values were compared with laboratory test results for the sand used in the experiment. The main parameters of soil properties are summarized in Table 9. The estimated parameters were derived through model updating using the surrogate model and field measurements. The comparison reveals that the estimated values exhibit good agreement with the experimentally measured parameters. The relative errors for all five parameters remain within acceptable bounds, with the largest deviation observed in Poisson's ratio (3.103 %) and the smallest in density (0.267 %). Notably, the friction angle and cohesion, which are critical parameters influencing shear strength and deformation characteristics in sandy soils, show minimal errors of 3.537 % and 0.594 %, respectively. These results confirm the effectiveness of the back analysis framework in estimating soil properties with high accuracy, even under conditions with limited measurement data. The close match also

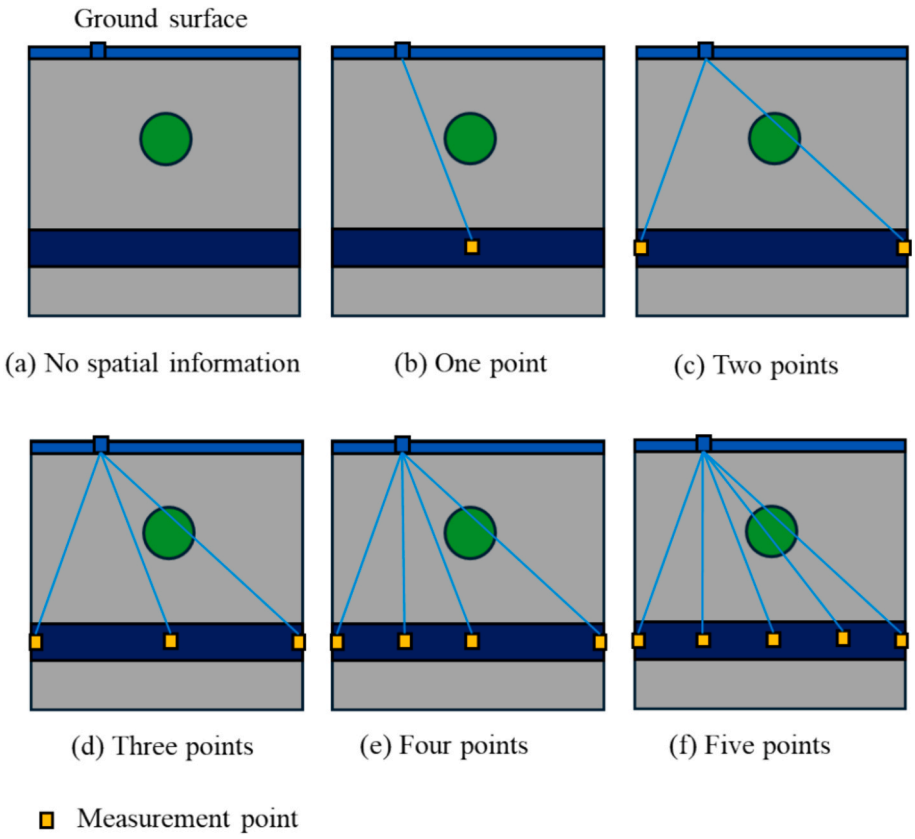


Fig. 11. Configurations of spatial measurement points used in the surrogate model.

Table 6
Effect of spatial information on ground settlement prediction accuracy.

Spatial information	MSE	RMSE	MAE	R2	MAPE
No point	1.7626	1.3276	1.4969	0.9704	1.7023
One point	1.2048	1.0976	0.9284	0.9879	1.1519
Two points	1.0482	1.0238	0.8580	0.9888	1.0728
Three points	0.8565	0.9254	**0.7102**	**0.9898**	**0.9215**
Four points	0.9040	0.9508	0.7443	0.9888	0.9508
Five points	**0.8530**	**0.9235**	0.7262	0.9891	0.9460

Table 7
Comparison of network architectures for ground settlement prediction.

Networks	MSE	RMSE	MAE	R2	MAPE
CNN-FCNN	1.6322	1.2776	0.8990	0.9911	1.5022
ResNet-FCNN	1.7370	1.3180	0.8703	0.9905	0.9745
ResNet-FiLM-I	1.0812	1.0398	0.6573	0.9941	0.9360
ResNet-FiLM-II	1.0336	1.0167	0.7418	**0.9944**	0.9418
Swin-T-I	1.2230	1.1059	0.7392	0.9933	0.9223
Swin-T-II	1.3385	1.1569	0.7437	0.9927	1.5429
DF-STTN	**0.8565**	**0.9254**	**0.7102**	0.9898	**0.9215**

supports the use of the updated surrogate model for further predictive tasks in similar geotechnical settings.

The effectiveness of the estimated soil parameters obtained through back analysis was further evaluated by comparing the resulting ground settlement predictions with both simulation outputs and field measurements. Fig. 13(a) presents the comparison along the horizontal direction, while Fig. 13(b) shows the results along the vertical axis. In both cases, the predicted settlements, computed using the updated soil properties, demonstrate closer agreement with the test data than the original numerical simulation results. In the horizontal direction, the updated model notably reduces the overestimation observed in the simulation. Along the vertical direction, the predicted results better capture both the trend and magnitude of the observed settlement. These findings confirm that the updated model enhances predictive accuracy. This updated model serves as a new surrogate model, which can be applied to future prediction tasks under similar or new configurations.

3.3. Surrogate model of lining deformation

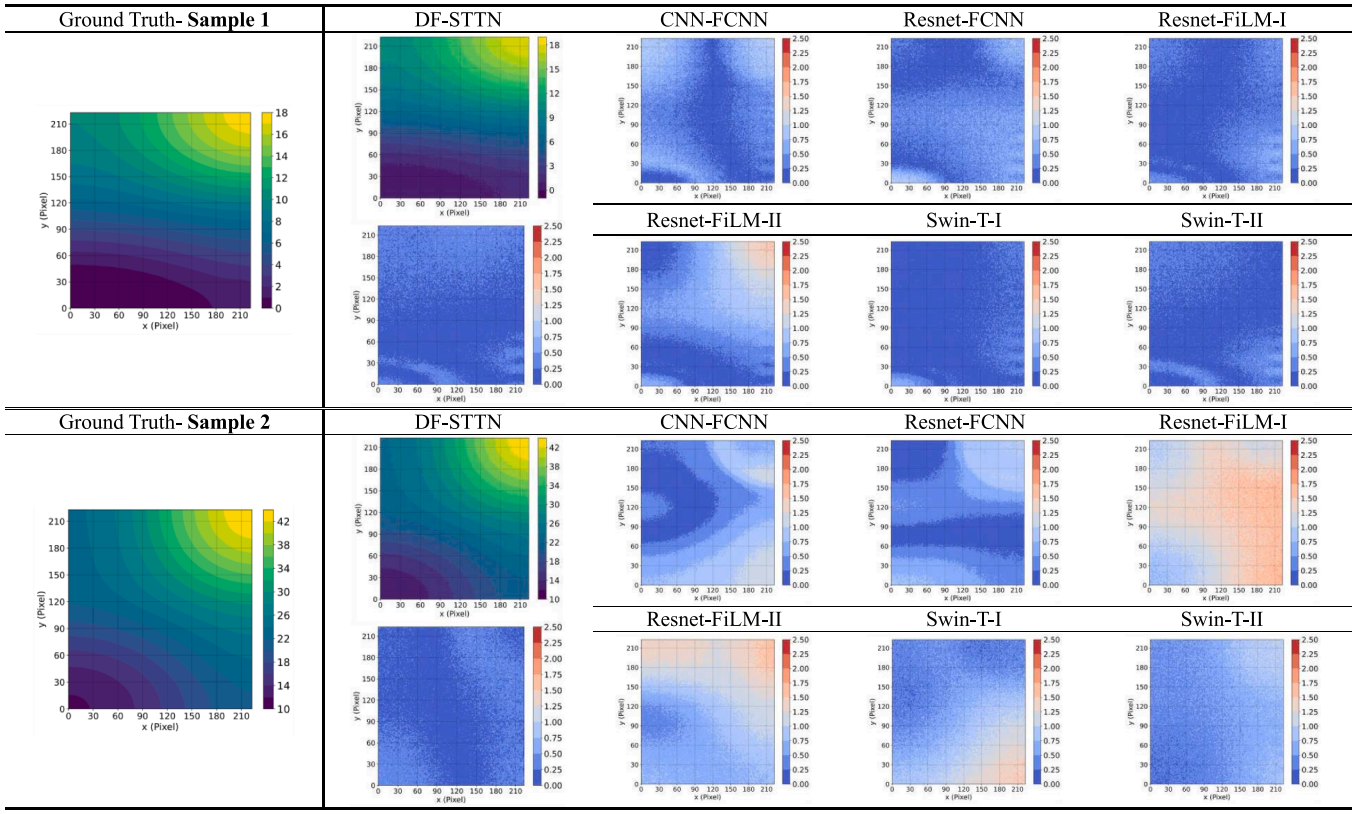
3.3.1. Surrogate model development

The influence of training sample size on model accuracy was assessed by increasing the number of training samples from 100 to 800. As shown in Table 10, the MSE value decreased rapidly from 0.0908 (100 samples) to 0.0074 (800 samples), indicating that additional training data substantially improved the model predictive capability. Similar to the ground settlement task, the model exhibited significant performance improvements with smaller sample sizes, while additional gains became progressively limited beyond 700 samples.

To evaluate the role of spatial input, models were trained using zero to five spatial measurement points (see Table 11). The results show that spatial information improved prediction accuracy across all metrics. The best performance was achieved using three spatial points (MSE = 0.0074, R^2 = 0.9951). Adding more than three points provided only modest additional gains, suggesting diminishing returns beyond this level. Overall, these results indicate that incorporating a small number of spatial points is sufficient to achieve high predictive accuracy while maintaining computational efficiency, highlighting the important role of spatial information in full-field deformation prediction.

Seven network architectures were tested for lining deformation prediction. As summarized in Table 12, the DF-STTN outperformed all other models, achieving the lowest MSE (0.0074), RMSE (0.0860), and MAE (0.0612), alongside the highest R^2 (0.9951). Transformer-based models generally outperformed convolution-based ones, particularly in capturing complex spatial patterns. Error maps for representative

Table 8
Ground settlement prediction and error maps from different models.



Note: The x and y axes represent pixel indices corresponding to different locations on the ground surface. The legend represents ground settlement (ground truth) and error values (error maps), and its unit is millimeters (mm).

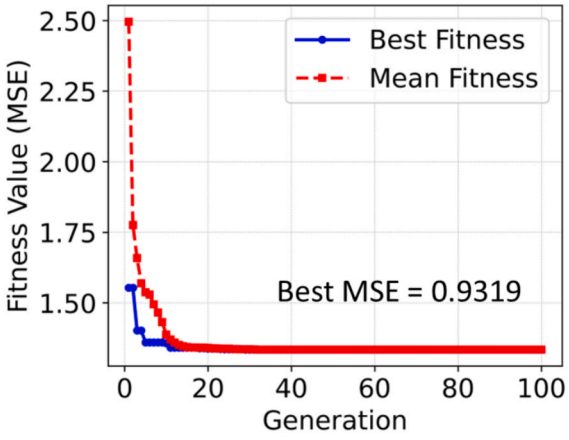


Fig. 12. GA convergence process during surrogate model updating for ground settlement.

samples (Table 13) further confirm these findings. The DF-STTN model produced the most accurate and smoothest predictions, closely matching the ground truth. Other models, such as CNN-FCNN and ResNet variants, exhibited higher residuals and spatial inconsistency. These results underscore the robustness and effectiveness of the DF-STTN model in modeling complex deformation behavior.

3.3.2. Model updating and analysis

To improve the surrogate model for lining deformation, the model updating procedure similar to that in Section 3.2.2 was implemented.

Table 9
Measured vs. predicted soil parameters for the ground settlement surrogate model.

Soil properties	Density (g/cm^3)	Elastic modulus (MPa)	Poisson's ratio	Friction angle (∘)	Cohesion (kPa)
Real value	1983.00	68.000	0.290	21.49	0.842
Estimated value	1977.69	69.573	0.281	22.25	0.847
Error (%)	0.267	2.313	3.103	3.537	0.594

The goal was to estimate the underlying soil parameters that best reproduce observed lining deformation at selected monitoring points. Due to the limited availability of test data, the back analysis focused on four key locations along the tunnel lining, as shown in Fig. 6. As illustrated in Fig. 14, both the best and mean fitness values (MSE) rapidly converged within the first 20 generations, indicating stable and effective optimization. The best fitness value reached 0.00098, demonstrating excellent agreement between predicted and test values.

The accuracy of the estimated soil parameters for the lining deformation model was assessed by comparing them against experimentally obtained values, as summarized in Table 14. The estimated parameters obtained through back analysis demonstrate high consistency with the real values, with relative errors all remaining below 4 %. Notably, cohesion shows the highest deviation (3.919 %), while elastic modulus and Poisson's ratio exhibit minor errors of 1.029 % and 1.379 %, respectively. Further validation is presented in Fig. 15, which compares the predicted lining deformation with both simulation data and test measurements along the tunnel horizontal direction. The predicted

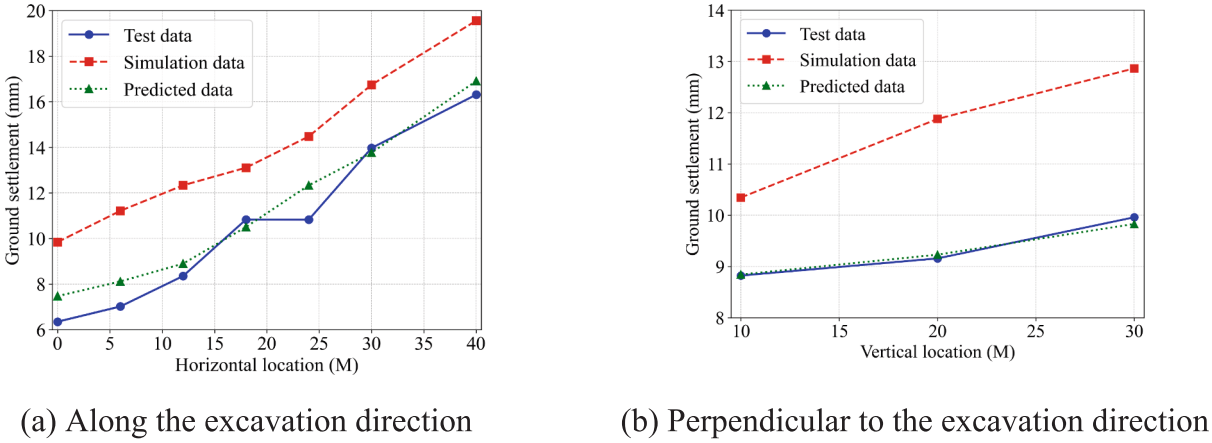


Fig. 13. Comparison of ground settlement data: test data vs. simulation and prediction results.

Table 10
MSE for Lining deformation prediction using different sample sizes.

Sample number	MSE
100 samples	0.0908
200 samples	0.0486
300 samples	0.0220
400 samples	0.0180
500 samples	0.0167
600 samples	0.0179
700 samples	0.0101
800 samples	0.0092

Table 11
Effect of spatial information on lining deformation prediction accuracy.

Spatial information	MSE	RMSE	MAE	R2	MAPE
No point	0.0190	0.1379	0.0929	0.9875	0.0589
One point	0.0148	0.1216	0.0873	0.9903	0.0564
Two points	0.0131	0.1144	0.0776	0.9914	0.0571
Three points	**0.0074**	**0.0860**	**0.0642**	**0.9951**	**0.0474**
Four points	0.0082	0.0905	0.0700	0.9948	0.0495
Five points	0.0084	0.0916	0.0682	0.9945	0.0481

Table 12
Comparison of network architectures for lining deformation prediction.

Networks	MSE	RMSE	MAE	R2	MAPE
CNN-FCNN	0.0151	0.1230	0.0881	0.9900	0.0558
ResNet-FCNN	0.0120	0.1093	0.0767	0.9921	0.0503
ResNet-FiLM-I	0.0080	0.0894	0.0619	0.9947	0.0454
ResNet-FiLM-II	0.0079	0.0889	0.0649	0.9948	0.0455
Swin-T-I	0.0099	0.0996	0.0690	0.9935	**0.0438**
Swin-T-II	0.0103	0.1017	0.0675	0.9932	0.0470
DF-STTN	**0.0074**	**0.0860**	**0.0612**	**0.9951**	0.0474

values based on the updated model align closely with the test data. The deviations are minimal across all positions, demonstrating that the updated surrogate model not only improves parameter estimation but also enhances predictive capability for deformation.

3.4. Rapid deformation prediction

To assess the generalization performance of the updated surrogate model, two test configurations, referred to as Tests 2 and 3, were introduced (see Fig. 7). The updated surrogate model directly predicted responses based on the estimated soil parameters obtained previously. Ground settlement predictions along both horizontal and vertical directions are shown in Fig. 16, respectively. The predicted values exhibit excellent agreement with the test data, accurately capturing both the trend and magnitude of settlement. Slight deviations appear at a few points, but the overall predictive quality remains high. Fig. 17 illustrates the predicted tunnel lining deformation for Tests 2 and 3. The predicted curve aligns closely with the measured deformation, including the peak location and deformation gradient across the tunnel section. This demonstrates the potential of the proposed framework as a key component of a DT system for tunnel engineering. In practical applications, DTs can be used to optimize tunnel design by simulating deformation responses under varying geometric layouts and construction parameters, enabling engineers to evaluate and select more resilient and cost-effective design alternatives during the planning phase.

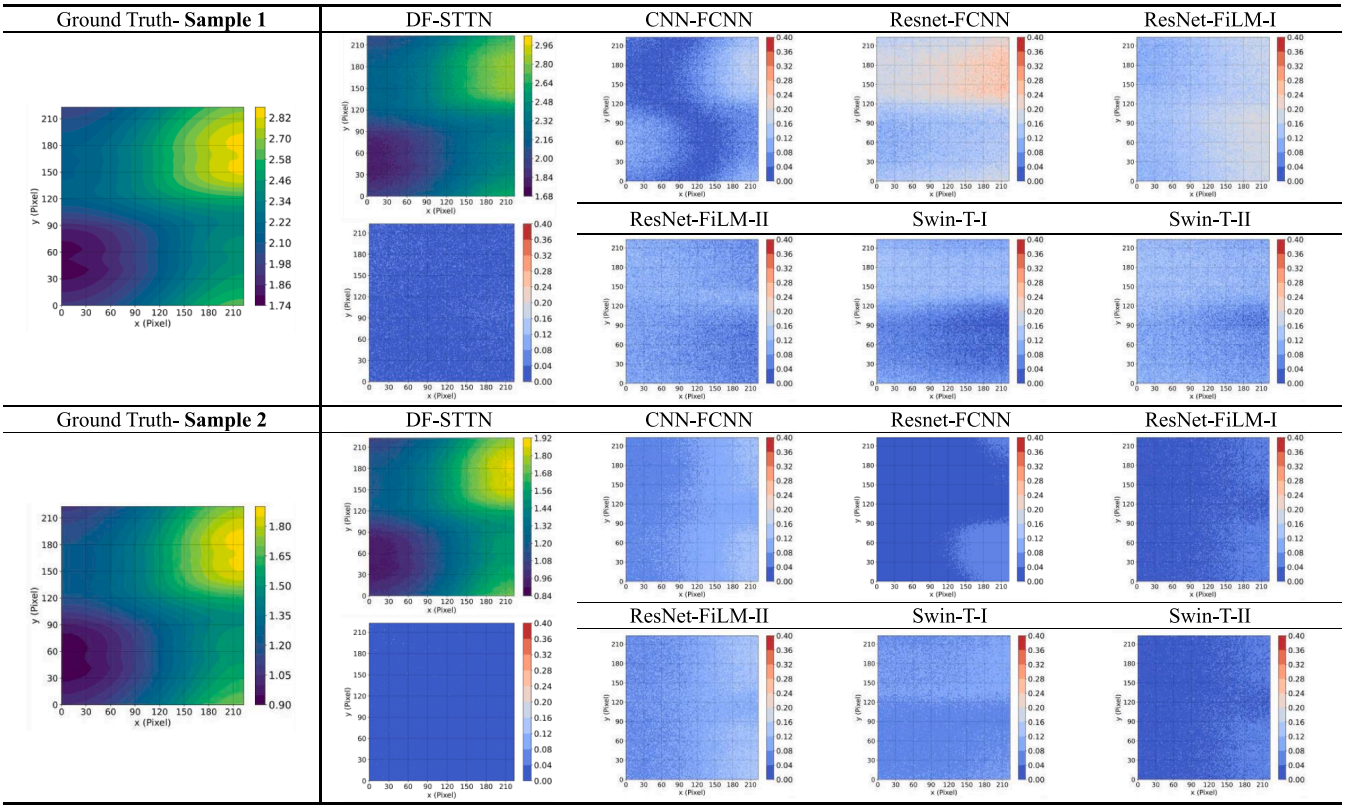
Overall, the successful application of the updated model to the Tests 2 and 3 scenario demonstrates its potential for rapid deployment in geotechnical practice. It enables efficient assessment of tunnel performance in different design setups without retraining or costly numerical simulations, greatly improving the practicality of data-driven modeling in underground engineering.

4. Conclusion

This study presents a digital twin framework for predicting ground settlement and shield tunnel deformation during adjacent tunnel construction. The proposed framework integrates a hybrid surrogate modeling approach, Dual-Fusion Swin-Table Transformer Network (DF-STTN), with model updating strategies to enable accurate and efficient simulations of tunnel behavior under varying design and soil conditions. By decoupling spatial and non-spatial inputs and leveraging advanced Transformer architectures, the model effectively captures complex spatial dependencies and tabular relationships. The following conclusions are obtained:

(1) The inclusion of spatial information was found to improve ground settlement prediction accuracy, with noticeable performance improvements observed even when three spatial measurement points were used. This highlights that limited spatial context can meaningfully enhance the modeling of excavation-induced responses.
(2) Comprehensive evaluations based on physical tunnel excavation tests demonstrate that DF-STTN outperforms conventional convolution-based and single-fusion models in both ground settlement and lining deformation tasks.
(3) A model updating procedure using genetic algorithms enables adaptive refinement of the surrogate model using sparse test data, ensuring consistency with observed field behavior.
(4) The updated model exhibits reliable generalization when applied to new configurations without additional retraining,

Table 13
Lining deformation prediction and error maps from different models.



Note: The x and y axes represent pixel indices corresponding to different locations on the tunnel lining. The legend represents lining deformation (ground truth) and error values (error maps), and its unit is millimeters (mm).

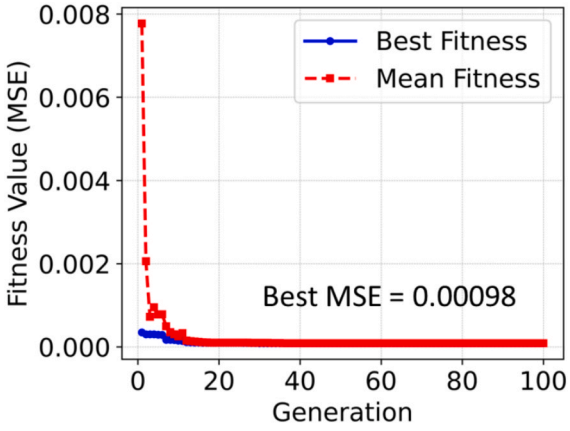


Fig. 14. GA convergence process during surrogate model updating for lining deformation.

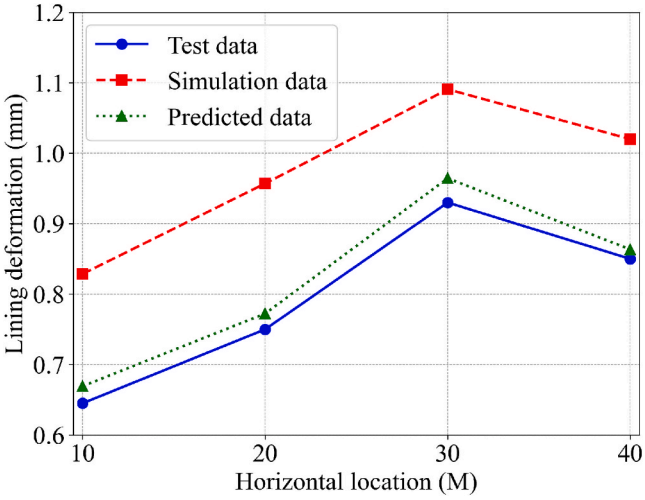


Fig. 15. Comparison of lining deformation data: test data vs. simulation and prediction results.

Table 14
Measured vs. predicted soil parameters for the lining deformation surrogate model.

Soil properties	Density (g/cm^3)	Elastic modulus (MPa)	Poisson's ratio	Friction angle (∘)	Cohesion (kPa)
Real value	1983.00	68.000	0.290	21.49	0.842
Estimated value	1973.62	67.300	0.286	20.73	0.875
Error (%)	0.473	1.029	1.379	3.537	3.919

underscoring its potential for rapid deployment in early-stage tunnel design.

By providing full-field predictions of ground settlement and tunnel lining deformation, the method allows engineers to assess potential risks more comprehensively than traditional point-based approaches. Its computational efficiency makes it feasible for use during the early

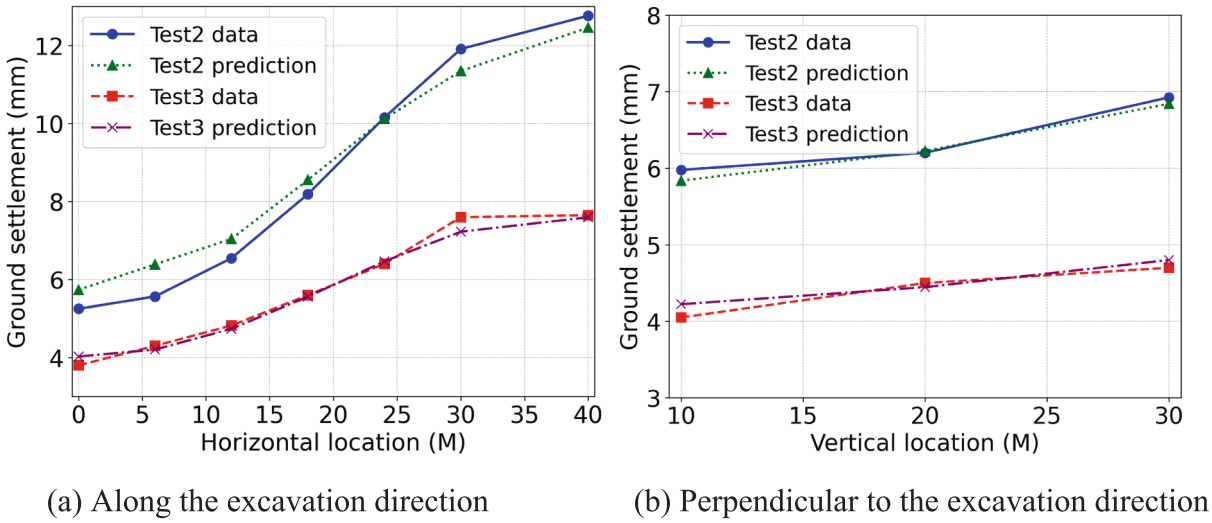


Fig. 16. Comparison of ground settlement data: test data vs. prediction results.

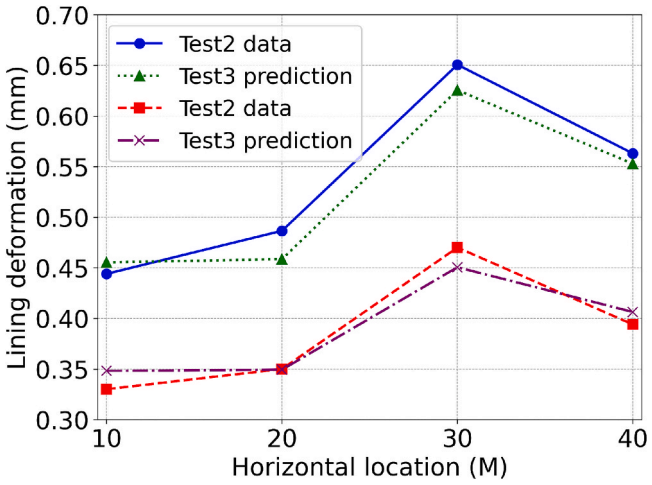


Fig. 17. Comparison of lining deformation data: test data vs. prediction results.

design stage, where multiple excavation scenarios and tunnel layouts must be evaluated rapidly. Furthermore, the framework supports adaptive updating with monitoring data, enabling continuous refinement of soil parameters and deformation predictions as new information becomes available. These features make the method a practical decision-support tool with guiding significance for adjacent tunnel construction projects.

Despite the promising results, several limitations should be acknowledged. First, validation was conducted using scaled laboratory tests, which, while controlled and repeatable, may not fully capture field-scale soil and structural responses; incorporating monitoring data in future work will help address this gap. Second, the current model focuses on spatial deformation prediction across excavation stages but does not explicitly capture time-dependent processes such as soil consolidation or long-term ground–structure interaction. Extending the surrogate model to incorporate temporal dynamics through spatio-temporal modeling is an important direction for future development. Third, ground settlement and lining deformation were modeled separately for interpretability and stability, but this neglects potential coupling effects; integrated models will be explored to capture these interactions.

Future work will focus on enhancing the framework to address more realistic and complex geotechnical conditions, including layered soil strata, groundwater influence, and nonlinear soil–tunnel interactions such as consolidation and creep. These factors are critical for accurately simulating deformation in soft ground tunneling projects. To improve modeling of sequential excavation processes, spatio-temporal Transformer architectures will be investigated to capture both temporal evolution and spatial dependencies in deformation patterns. Moreover, integration with real-time sensor data, leveraging technologies such as wireless sensor networks, automated total stations, and fiber-optic monitoring, will enable continuous model updating and adaptive feedback, supporting dynamic decision-making in a digital twin environment.

CRediT authorship contribution statement

Xiangyu Chang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Hongyun Fan:** Writing – review & editing, Software, Investigation, Data curation, Conceptualization. **Yuguang Fu:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization. **Chengjia Han:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Hao Wang:** Writing – review & editing, Supervision, Methodology, Data curation, Conceptualization. **Jianxiao Mao:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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